

Solving Nonlinear Inequalities (Rational Expressions) Homework

Solve. Write the solutions graphically and in interval notation.

$$\textcircled{1} \frac{2x^2}{9-x^2} + \frac{11x}{9-x^2} \geq \frac{3x}{9-x^2} - \frac{2x^2}{9-x^2}$$

$$\textcircled{2} \frac{36x^2}{5x^2+x-18} > \frac{1}{5x^2+x-18}$$

$$\textcircled{3} \frac{-3x^3}{x+4} + \frac{12x^2}{x+4} + \frac{36x}{x+4} \leq 0$$

$$\textcircled{4} \frac{x^4 + 15x^2 - 250}{x^3 - 6x^2 + 7x - 42} < 0$$

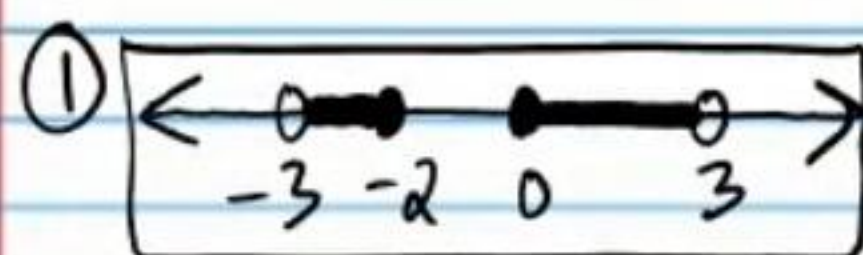
$$\textcircled{5} \quad 0 \leq \frac{x^2 + 6x + 8}{8x^3 + 125}$$

$$\textcircled{6} \quad \frac{x-7}{x-5} \leq 2$$

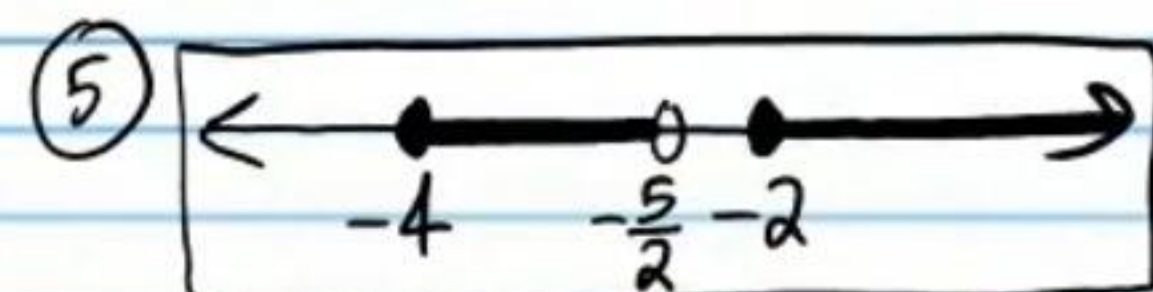
$$\textcircled{7} \quad -1 + \frac{6}{x} > x$$

$$\textcircled{8} \frac{x-7}{x-4} < \frac{-4}{x-3}$$

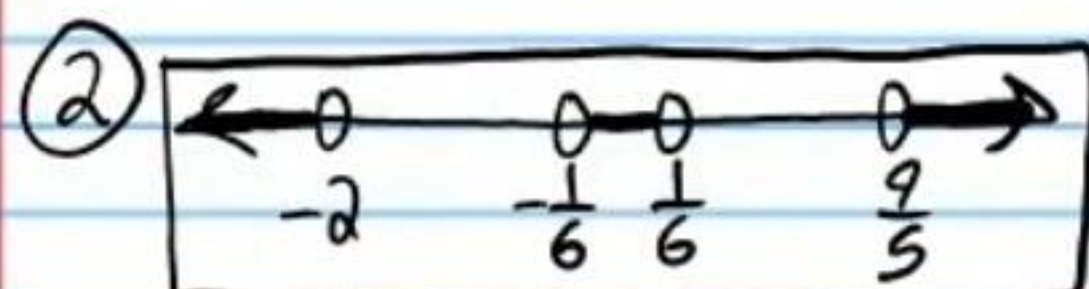
Answers



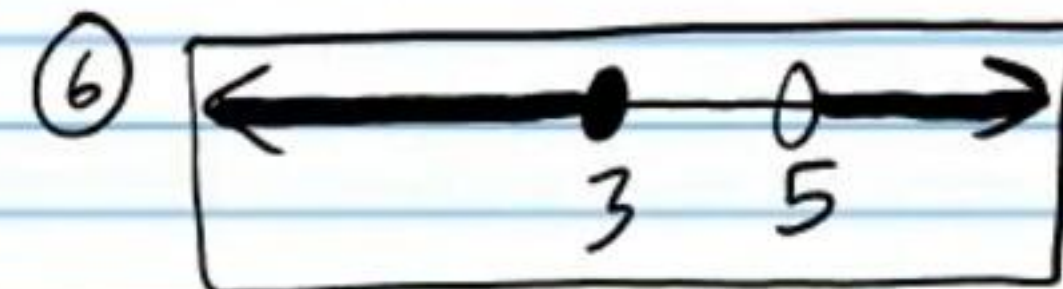
$$(-3, -2] \cup [0, 3)$$



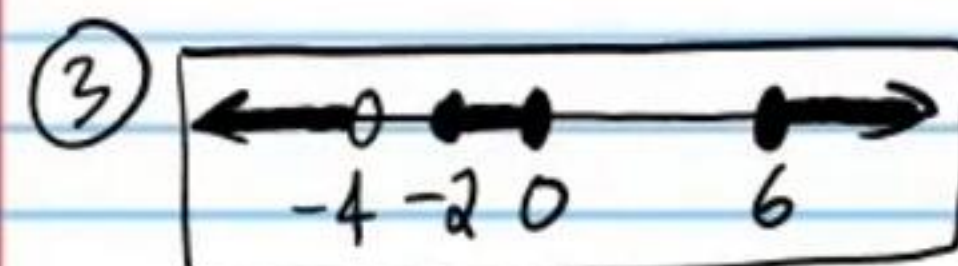
$$[-4, -\frac{5}{2}) \cup [-2, \infty)$$



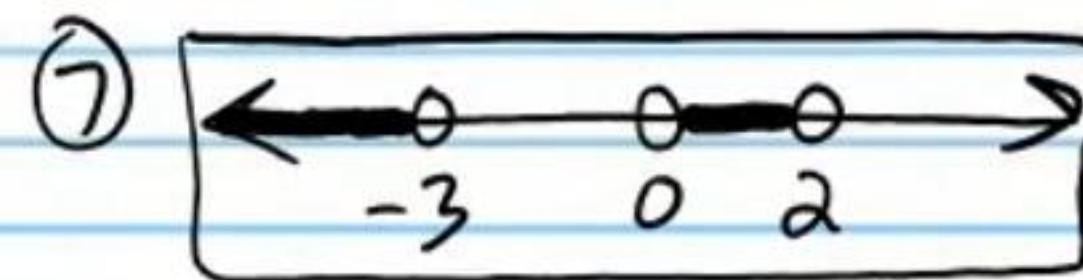
$$(-\infty, -2) \cup (-\frac{1}{6}, \frac{1}{6}) \cup (\frac{2}{5}, \infty)$$



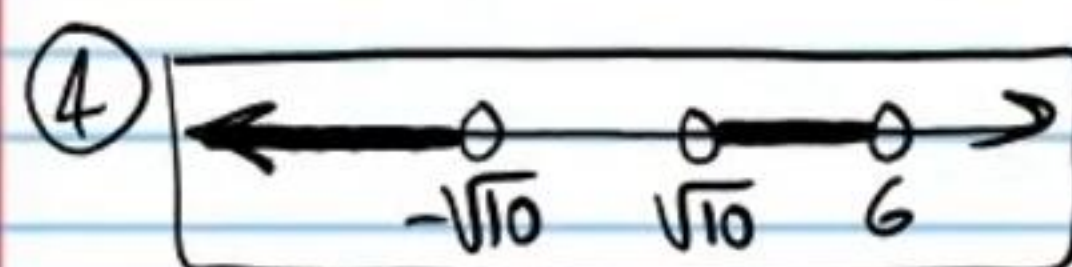
$$(-\infty, 3] \cup (5, \infty)$$



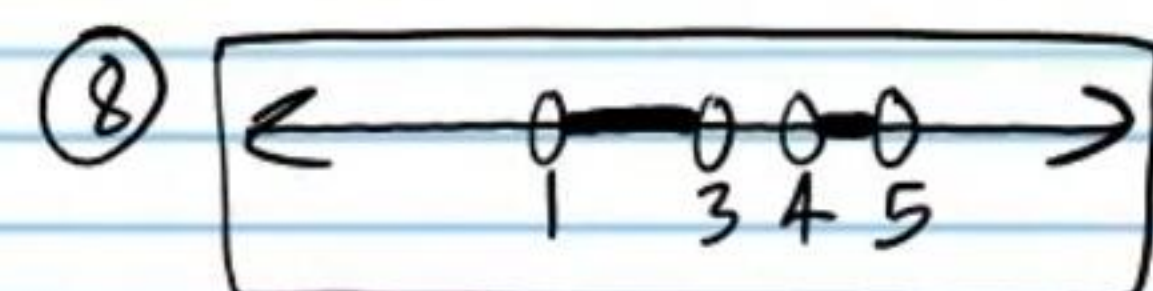
$$(-\infty, -4) \cup [-2, 0] \cup [6, \infty)$$



$$(-\infty, -3) \cup (0, 2)$$



$$(-\infty, -\sqrt{10}) \cup (\sqrt{10}, 6)$$



$$(1, 3) \cup (4, 5)$$